ARTIFICIAL INTELLIGENCE

How AI is changing our world

AI Subject Presentation | Information Technology

Presented by: ___Gohil Riya Rajendrasinh _____Roll No.: _____26IT050_____



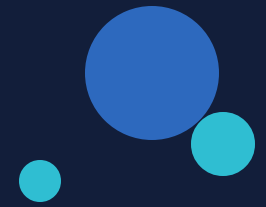
1. What Is Artificial Intelligence?

- Artificial Intelligence (AI) is a branch of computer science.
- It helps machines perform tasks that usually need human intelligence.
- AI systems can learn from data, recognize patterns, and make predictions.
- Examples include voice assistants, chatbots, and face recognition.

AI = Data + Algorithms + Computing

2. How Does AI Work?

- Input: information is collected from text, images, sensors, or users.
- Processing: algorithms analyze the input and find patterns.
- Learning: some AI systems improve by training on examples.
- Output: the system gives a prediction, recommendation, or response.
- The quality of results depends on data, design, and testing.



INPUT → PROCESSING → OUTPUT

3. Types of AI

- Narrow AI (ANI): designed for specific tasks; used today in many products.
- General AI (AGI): a theoretical AI with broad human-like abilities.
- Super AI (ASI): a hypothetical system that would exceed human abilities.
- Most real-world AI today is Narrow AI.

TODAY:
Mostly
Narrow AI

4. AI in Everyday Life

- Smartphones: voice typing, camera enhancement, and face unlock.
- Entertainment: movie, music, and video recommendations.
- Navigation: traffic estimates and route suggestions.
- Online services: spam filters, chatbots, and product recommendations.
- Smart devices: assistants and automated home features.

AI is already part of daily life

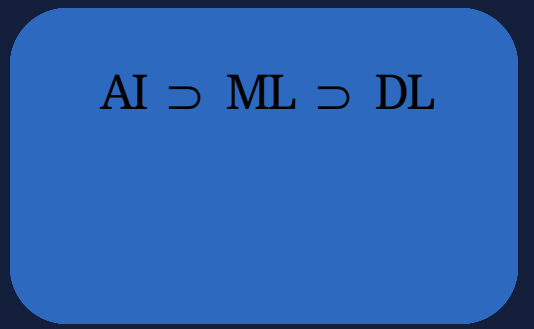
5. AI in Healthcare and Education

- Healthcare: helps analyze scans and supports clinical decision-making.
- Healthcare: can help monitor trends, but professionals make care decisions.
- Education: provides practice questions and personalized learning support.
- Teachers and healthcare workers remain important for judgment and care.
- AI outputs should be checked for accuracy and bias.

AI SUPPORTS PEOPLE

6. Machine Learning and Deep Learning

- Machine Learning (ML) is a part of AI that learns patterns from data.
- Deep Learning (DL) is a type of ML using multi-layer neural networks.
- Example: ML can classify messages as spam or not spam.
- Example: DL can help recognize objects in photos and understand speech.
- AI is the broad field; ML and DL are related subfields.



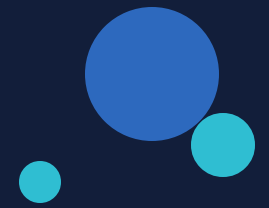
7. Generative AI

- Generative AI creates new content based on patterns learned from data.
- It can generate text, images, audio, video, and computer code.
- Examples include AI writing assistants, image generators, and coding tools.
- It can be useful for brainstorming, learning, and creating first drafts.
- Generated content may contain errors and should be reviewed.

CREATE •
CHECK •
IMPROVE

8. Benefits and Challenges

- Benefits: automates repetitive tasks and can save time.
- Benefits: helps analyze large datasets and supports accessibility.
- Challenges: biased data can lead to unfair results.
- Challenges: privacy, misinformation, security, and job changes need attention.
- Responsible AI needs transparency, human oversight, and careful testing.



USE AI RESPONSIBLY

9. Conclusion

- AI enables computers to perform tasks associated with human intelligence.
- It is used in communication, education, healthcare, transport, and business.
- ML and DL help power many modern AI applications.
- AI is a useful tool, but it is not perfect or always correct.
- The best results come from combining AI with human judgment.

THANK YOU!

Any Questions?